

11.4-Logistic_Regression_With_ROC_and_AUC

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1 Logistic Regression with ROC And AUC Score

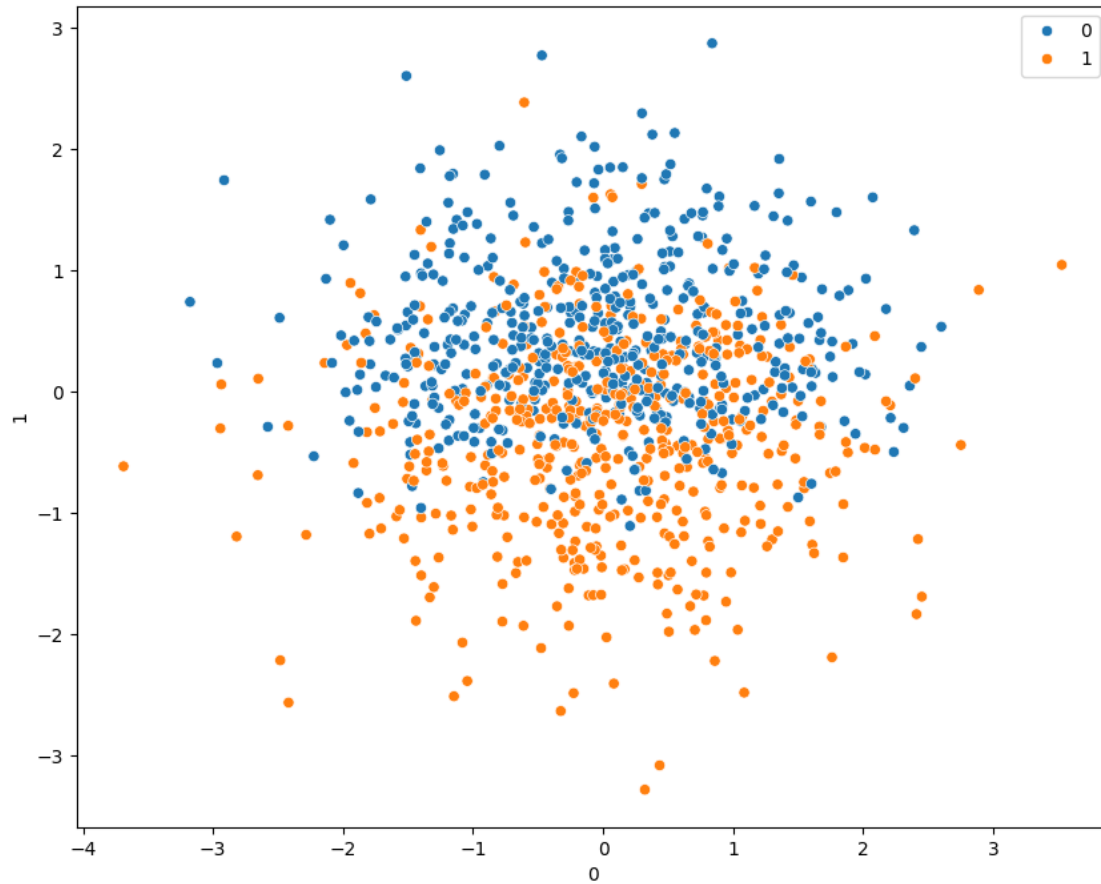
```
[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_curve, roc_auc_score
```

```
[2]: ## generating 2 class dataset
X, y = make_classification(n_samples=1000, n_classes=2, random_state=42)
```

```
[3]: pd.DataFrame(y).value_counts()
```

```
[3]: 0
0    500
1    500
Name: count, dtype: int64
```

```
[4]: plt.figure(figsize=(10,8))
sns.scatterplot(x=pd.DataFrame(X)[0], y=pd.DataFrame(X)[1], hue=y)
plt.show()
```



```
[5]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
↳ random_state=42)
```

```
[6]: # we have a dummy model with default 0 as output (majority class)
dummy_model_prob = [0 for _ in range(len(y_test))]
# dummy_model_prob
```

```
[7]: model = LogisticRegression()
model.fit(X_train, y_train)
```

```
[7]: LogisticRegression()
```

```
[8]: model_prob = model.predict_proba(X_test)
```

```
[9]: model_prob = model_prob[:, 1] ## taking the positive value probability
```

```
[10]: dummy_model_auc = roc_auc_score(y_test, dummy_model_prob)
model_auc = roc_auc_score(y_test, model_prob)
print(dummy_model_auc)
```

```
print(model_auc)
```

0.5

0.9141713014460511

1.1 Calculating ROC Curves

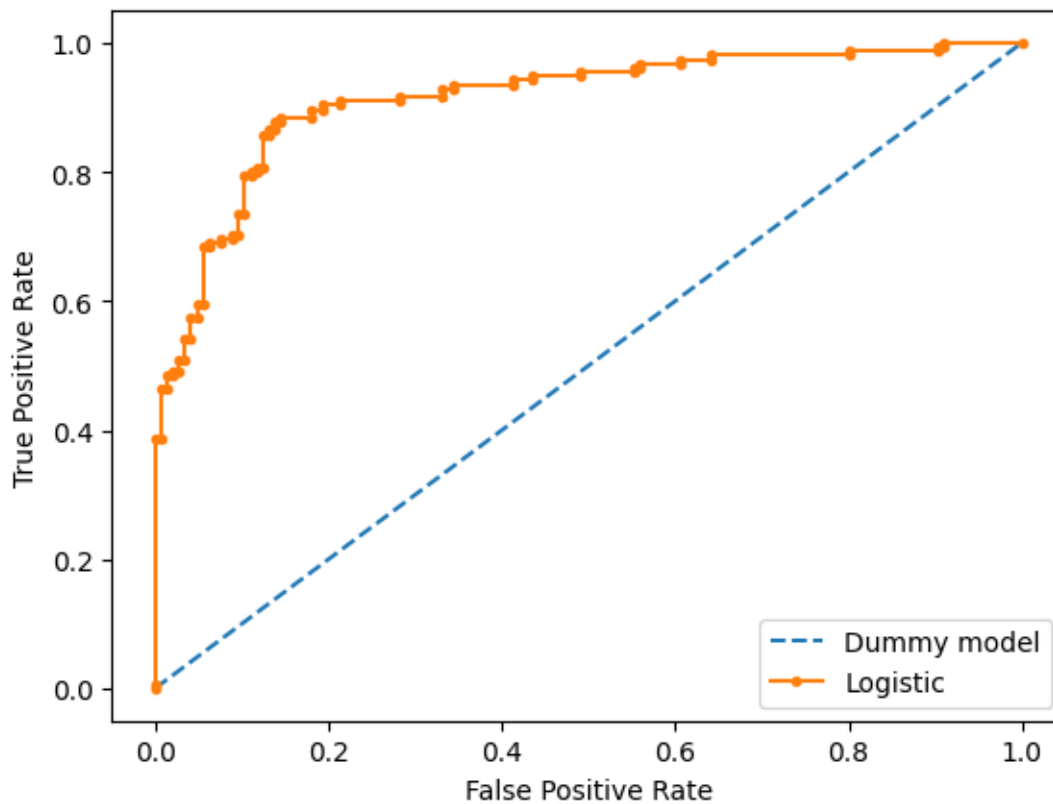
```
[11]: dummy_fpr, dummy_tpr, _ = roc_curve(y_test, dummy_model_prob)
      model_fpr, model_tpr, threshold = roc_curve(y_test, model_prob)
```

2 ROC curve

```
[12]: plt.plot(dummy_fpr, dummy_tpr, linestyle = '--', label = 'Dummy model')
      plt.plot(model_fpr, model_tpr, marker = '.', label = 'Logistic')

      plt.xlabel('False Positive Rate')
      plt.ylabel('True Positive Rate')

      plt.legend()
      plt.show()
```



```
[13]: threshold
```

```
[13]: array([          inf,  9.99903312e-01,  9.60819585e-01,  9.60673656e-01,
          9.42928905e-01,  9.37634035e-01,  9.35829937e-01,  9.35196941e-01,
          9.33388706e-01,  9.31568682e-01,  9.19898462e-01,  9.18320734e-01,
          9.08848834e-01,  9.04149376e-01,  8.68929260e-01,  8.68816110e-01,
          8.59454363e-01,  8.57330834e-01,  7.66609556e-01,  7.56307961e-01,
          7.47504157e-01,  7.44341554e-01,  7.38336625e-01,  7.35803961e-01,
          7.28263642e-01,  7.19672711e-01,  6.70517119e-01,  6.64080329e-01,
          5.97992204e-01,  5.63115004e-01,  5.61146495e-01,  5.42332260e-01,
          5.33104469e-01,  5.18849321e-01,  4.67721242e-01,  4.66279420e-01,
          4.53793669e-01,  4.43620311e-01,  4.30357018e-01,  4.30169935e-01,
          4.29200970e-01,  3.96059977e-01,  3.47982821e-01,  3.34198768e-01,
          3.28119070e-01,  3.07874692e-01,  2.95655789e-01,  2.40142099e-01,
          2.39704326e-01,  1.97988953e-01,  1.92236077e-01,  1.72072589e-01,
          1.68324514e-01,  1.44381585e-01,  1.41194398e-01,  1.36275906e-01,
          1.36153807e-01,  1.15365849e-01,  1.09775834e-01,  8.42967572e-02,
          8.34570647e-02,  8.06346852e-02,  7.91753836e-02,  6.79806651e-02,
          6.57812599e-02,  5.99608153e-02,  5.97147719e-02,  3.54892498e-02,
          3.28958636e-02,  1.18188476e-02,  1.14111255e-02,  1.00222414e-02,
          9.13342692e-03,  6.15544334e-05])
```

2.1 All The Threshold Values

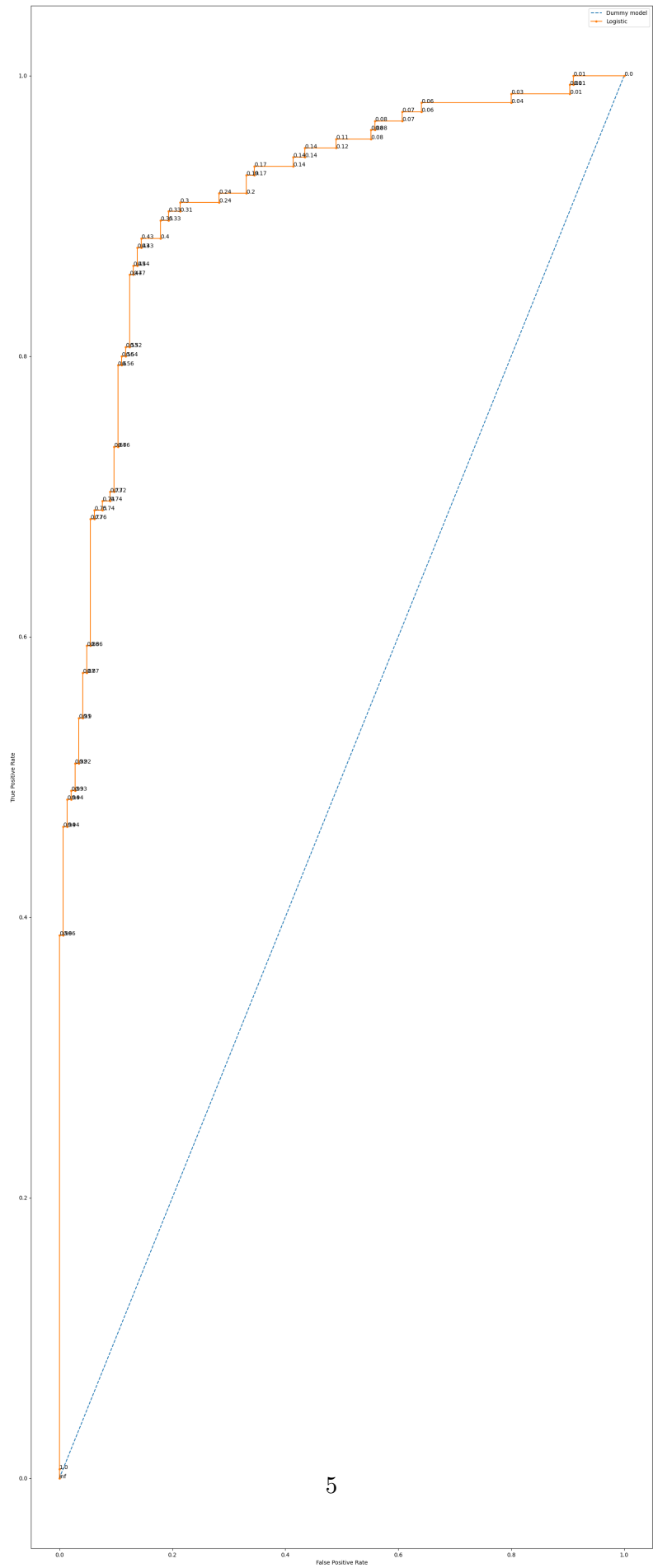
```
[14]: # Create figure and axis FIRST
fig = plt.figure(figsize=(20, 50))
ax = fig.add_subplot(111)

ax.plot(dummy_fpr, dummy_tpr, linestyle='--', label='Dummy model')
ax.plot(model_fpr, model_tpr, marker='.', label='Logistic')

for xyz in zip(model_fpr, model_tpr, threshold):
    ax.annotate('%s' % np.round(xyz[2], 2), xy=(xyz[0], xyz[1]))

ax.set_xlabel('False Positive Rate')
ax.set_ylabel('True Positive Rate')
ax.legend()

plt.show()
```



```

[15]: from sklearn.metrics import roc_curve, auc

dummy_fpr, dummy_tpr, _ = roc_curve(y_test, dummy_model_prob)
model_fpr, model_tpr, threshold = roc_curve(y_test, model_prob)

dummy_auc = auc(dummy_fpr, dummy_tpr)
model_auc = auc(model_fpr, model_tpr)

# Create plot
plt.figure(figsize=(10, 8))

# Plot ROC curves
plt.plot(dummy_fpr, dummy_tpr, linestyle='--', label=f'Dummy Model (AUC = {dummy_auc:.2f})')
plt.plot(model_fpr, model_tpr, marker='.', label=f'Logistic Model (AUC = {model_auc:.2f})')

for i in range(0, len(threshold), 10):
    plt.annotate(
        text=f"{threshold[i]:.2f}",
        xy=(model_fpr[i], model_tpr[i]),
        xytext=(5, 5),
        textcoords='offset points'
    )

plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve Comparison')
plt.legend(loc='lower right')
plt.grid(True)
plt.show()

```

